

Technology News

Walk-assist robot for the disabled

Toyota has unveiled a walk-assist robot that can be worn by people who have walking disability due to leg paralysis or other causes. The robot is attached to one leg and helps to walk and get in or out of the bed.



Walk Assist robot by Toyota

The Walk Assist robot employs intelligent multi-sensor control—a position sensor on the thigh and a pressure sensor on the sole of the foot, which detect the user's thigh position and the weight put on the foot. Based on the user's stride, an actuator at the knee regulates action to achieve the best walking assistance. By calculating the walking speed, the user is able to walk naturally. A special link mechanism at the knee joint ensures a secure lock (without the knee bending) when the leg swings and plants the weight-bearing foot on the ground, so that the user can walk with confidence.

Thinking chip

As part of a cognitive computing project called Systems of Neuromorphic

Adaptive Plastic Scalable Electronics (SyNAPSE), IBM researchers have unveiled computer chips designed to emulate the brain's abilities for perception, action and cognition. The chips use digital silicon circuits inspired by neurobiology to make up what is referred to as a 'neurosynaptic core' with integrated memory (replicated synapses), computation (replicated neurons) and communication (replicated axons).

IBM has two working prototype designs. Both cores were fabricated in 45nm silicon-on-insulator complementary metal-oxide semiconductor and contain 256 neurons. One core contains 262,144 programmable synapses and the other contains 65,536 learning synapses.

Called cognitive computers, systems built with these chips won't be programmed the way today's traditional computers are. Rather, these are expected to learn through experiences, find correlations, create hypotheses, and remember and learn from the outcomes—mimicking the brain's structural and synaptic plasticity.

Video display screens turned into virtual whiteboards

Polycom has unveiled a whiteboard solution that improves communication and collaboration in video conferences by allowing meeting participants to leverage their existing LCD monitors and display screens of mobile devices as interactive whiteboards for video collaboration.

When traditional whiteboards are used in a video conference, remote participants have trouble seeing the whiteboard content. The Polycom UC Board solution combines a plug-and-play receiver and a stylus for ease-of-use, and a compact design for portability. The compact infrared sensor attaches to the whiteboard or LCD screen, and the wireless stylus can be used by the

meeting hosts to sketch, write and annotate just like any pen or marker.

The Polycom UC Board software natively integrates with the latest Polycom RealPresence Room HDX solutions to share content. Users participating in a video meeting can also share content such as a presentation.

Flexible, waterproof robot for rescue mission

Researchers at Chiba Institute of Technology in Japan have developed a robot that mimics the way a slug moves, and is wrapped in a flexible, waterproof 'skin' for search-and-rescue operations. The Slug Crawler Vehicle (SCV), enveloped in a retractable exterior skin layer, is protected from contaminants like dust and water that can corrode or damage the system in wet environments. The robot can climb up little slopes and over obstacles with relative ease.

Two internal pulleys swivel from side to side to manipulate the tread-like skin and steer the SCV. Together, these pull the tread into an arc shape to change the direction of movement.

Smartphone cardiac monitor

Researchers at the Swiss university Ecole Polytechnique Fédérale de Lausanne (EPFL) have developed a non-invasive wireless cardiac monitor app for smartphones. The app is intended to alert potential patients and their doctors to cardiac anomalies.

A wireless body sensor network is used, which consumes very little electricity. It is made up of four high-precision sensors attached to the skin, which are linked to a ZigBee radio module and a chip that clips onto the patient's belt to transmit ECG data to the smartphone app. The data can be viewed on the smartphone in real time. Complex algorithms flag up any abnormalities with data sent to a doctor for examination via a picture